Survey sampling weights in machine learning

"Use of survey sampling weights in machine learning models trained on Demographic and Health Surveys or national household surveys. Return a matrix with: authors, year, journal, DOI, survey, how weights were used (training, evaluation or both), and effect on results. Limit to 2015-2026." *Generated by LeapSpace*

Survey weights were used in these DHS or national household survey ML studies in three main ways: **training only**, **evaluation only**, or **both training and evaluation**. Where authors compared weighted and unweighted approaches, the effect was usually a change in calibration, sensitivity/recall, or apparent predictive performance rather than a dramatic change in ranking of the best model [1] [2] [3].

Matrix

Authors	Year	Journal	DOI	Survey	How weights were used	Effect on results
Authors not provided in retrieved evidence	2023	*PLoS ONE*	10.1371/journal.pone.0280387	NHANES 1988-1994 cohort	**Evaluation**: unweighted gradient boosting was trained; performance was recalculated with weighted observed outcomes for comparison [1].	Unweighted F1 was inflated versus weighted F1 (81.9% vs 77.4%); post-hoc weighted evaluation reduced this inflation [1].
Authors not provided in retrieved evidence	2024	*Epidemiology*	10.1097/EDE.0000000000001776	National Survey on Drug Use and Health; National Comorbidity Survey	**Evaluation**: survey weights were used to estimate test-set sensitivity/specificity; models were also trained with upsampling for imbalance [2].	Unweighted test metrics misrepresented population performance; weighted metrics better reflected the complex design [2].
Authors not provided in retrieved evidence	2025	*Annals of Epidemiology*	10.1016/j.annepidem.2024.12.016	US NHANES	**Training and evaluation comparison**: ML models were run without sampling weights and compared to a design-based logistic model; resampling handled imbalance [4].	Predictive capacity was similar to survey-design methods overall, suggesting that for relatively homogeneous data, simpler sampling may suffice for prediction [4].
Authors not provided in retrieved evidence	2025	*Data*	10.3390/data11050103	2023 NSCH	**Comparison of weighted vs unweighted learning** [5].	Survey weighting did **not** improve discriminative performance, and the effect was modest and model-dependent [5].
Authors not provided in retrieved evidence	2025	*Scientific Reports*	10.1038/s41598-025-24446-1	NFHS-5 (India)	**Training**: SMOTE was applied on weighted NFHS data, explicitly noting the complex survey design [6].	Reported high performance for Random Forest (test AUC 0.991); the abstract does not isolate the causal effect of weighting, but states

						weights were used while handling class imbalance [6].
Authors not provided in retrieved evidence	2025	*Journal of Computational Social Science*	10.1007/s42001-022-00195-3	Household panel data, Malawi	**Training**: sample weights were embedded into the cost function of ML models [7].	Training with weights increased recall for LASSO and MNN by roughly 11-12 percentage points; weighted models also showed much higher recall than baseline [7].
Authors not provided in retrieved evidence	2026	*Discover Artificial Intelligence*	10.1007/s44163-026-02092-4	DHS pooled across Sub-Saharan Africa	**Not used in training**: weights were used in descriptive analyses only, not integrated into the ML models [8].	Authors state this may affect variance estimation and feature-importance interpretation; results should be read as unweighted prediction from pooled samples [8].
Authors not provided in retrieved evidence	2026	*Biodata Mining*	10.1186/s13040-025-00505-1	Ghana DHS	**Evaluation and inference**: sampling weights, PSU, and strata were used for descriptive statistics and regression inference; ML training could not fully incorporate them [9].	Survey design was only partially integrated; authors note this limits full design-consistent inference from the ML pipeline [9].
Authors not provided in retrieved evidence	2026	*BMC Pregnancy and Childbirth*	10.1186/s12884-026-09255-2	2022 Bangladesh DHS	**Training**: sample weights, stratification, and clustering were incorporated into all modeling steps via `sample_weight` and cluster-aware splitting [10].	Weighted, survey-aware XGBoost achieved the best calibration and discrimination (AUROC 0.828; PR-AUC not top line but strong), and the authors present this as a robust predictive framework [10].
Authors not provided in retrieved evidence	2026	*Intelligence-Based Medicine*	10.1016/j.ibmed.2026.100504	Somalia demographic survey	**Training and evaluation**: continuous survey weights were integrated into model fitting; performance was assessed on a hold-out set [11] [12].	Weighted sample represented a larger national population; random forest performed best, but authors caution that residual survey-design effects may still influence performance and feature rankings [11] [12].
Authors not provided in retrieved evidence	2026	*Public Health Challenges*	10.1002/puh2.70413	Pooled DHS data from 64 LMICs	**Evaluation design / modeling pipeline**: held-out testing with SMOTENC inside training folds; the abstract says more complete consideration of complex survey design is still needed [13].	Overall performance was modest; authors conclude that stronger validation and fuller design handling are needed before practical use [13].
Authors not provided in retrieved evidence	2026	*Healthcare (Switzerland)*	10.3390/healthcare14142144	NFHS-5	**Training and evaluation**: DHS sampling weights were applied in a leak-controlled, grouped cross-validation framework [3].	Survey-weighted HGB had the strongest performance, but PPV was low and state-level transportability varied, limiting deployment to screening [3].
Authors not provided in retrieved evidence	2026	*BMC Women's Health*	10.1186/s12905-026-04705-7	DHS from 11 East African countries	**Training and evaluation**: country-specific sampling weights,	CatBoost performed best (AUC 80.09%); weighted pooled DHS

					stratification, and clustering were applied; model transportability was tested with leave-one-country-out CV [14].	modeling gave moderate transportability and reliable calibration [14].
Authors not provided in retrieved evidence	2026	*International Journal of Gynecology and Obstetrics*	10.1002/ijgo.71414	DHS from 10 Sub-Saharan African countries	**Training and evaluation**: individual-level DHS sampling weights were incorporated into weighted logistic regression, elastic net, random forest, and XGBoost [15].	Weighted random forest showed good internal and external discrimination (AUROC 0.863 and 0.878); SHAP and weighted regression identified FGM and symptom history as key predictors [15].

Bottom line

Across these studies, **using survey weights most often mattered for evaluation and calibration, and sometimes improved recall or population representativeness, but the effect on raw discrimination was mixed**: some studies found clear inflation in unweighted metrics [1], others found little or no gain in discrimination [5], and several recent DHS-based papers used weights throughout the pipeline to preserve national representativeness and design-consistent inference [3] [10] [14] [15].

Confidence level: **High** — the retrieved studies directly document how weights were used and, in several cases, quantify the effect on model performance; the main limitation is that author names were not included in the provided evidence.

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